AudioNinjaScrolls

System Calibration Checklist & Step-by-Step Workflow

Reference Formulas

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Speed of sound (air):     c = 331 + 0.6×T(°C) → 343 m/s @ 20°C
Propagation delay:        Δt = d / c
Transfer function:         H(f) = Y(f) / X(f)
dB gain, power ratio:     dB = 10 log(Po/Pi)
dB gain, distance/pressure: dB = 20 log(Do/Di)
Inverse square law:      every doubling of distance = −6 dB
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⚠ *Use 343 m/s consistently at ~20°C. dBu = analog voltage reference only; use dBFS for digital levels — never mix the two.*

Step 1 — Target SPL & Headroom Mapping

1. Measure ambient noise — SPL meter, C-weighting, Slow response, at the reference position (e.g., mix position or middle of the room)
2. Play back pink noise with the system at unity gain (0 dBu analog / −18 dBFS digital)
3. Measure the pink noise playback
4. Adjust amplifier gain until SPL reads ambient + 40 dB (this builds in sufficient headroom)
5. Pull output faders down 20 dB to establish the nominal working level, reserving a 20 dB cushion for venue-population drop

Example: ambient = 60 dB SPL → unity pink noise reads 100 dB SPL → nominal working level = 80 dB SPL.

Coverage Mapping Targets

Zone	Target
Front row	≤ 100 dB SPL (or lower per noise floor)
Middle row	−6 dB SPL vs. front row
Back row	≥ +10 dB SPL above noise floor
Middle row drop > 10 dB	Deploy time-aligned fill — do NOT push FOH level

Step 2 — Empirical Time / Phase Alignment

Tools: RTA microphone, playback track, DAW (e.g., Reaper), phase correction plug-in, audio interface.

Revised: this step now carries the full waveform-matching procedure from the Time Alignment Technique in the AudioNinjaScrolls manuscript. All eight original lines are kept, in order of use. Repeat the whole pass for each cluster: kick for Sub to Main, snare for Front Fill and Delay.

Build the Reaper template (once, then reuse for every venue)

Source: the Reaper template build in Sound System Calibration (the Reaper demo version is all you need), checked against 00 Calibration Template.RPP. Items marked [template] are in the .RPP but not in the book.

- **T1.** **New project.** Sample rate 44.1 kHz or higher, bit depth 24-bit. Set the audio device: ASIO driver on Windows, the audio driver on Mac.
- **T2.** **Tones folder.** Create a track named Tones. Create a Playback track and drag it into Tones; drop in any music with a decent mix of frequency content.
- **T3.** **Pink Noise.** New track, drag into Tones, **mute it**, add a pink noise generator through the FX button. [template: JS Liteon pink noise generator]
- **T4.** **Sine Sweep.** New track, drag into Tones, **mute it**, add a sine sweep generator. [template: JS sine_sweep, fader pulled down about 21 dB]
- **T5.** **1K Tone** [template only]. Muted track in Tones with a tone generator at 1000 Hz, −12 dB, for level checks.
- **T6.** **RTA track.** New track for the RTA mic. Open routing and **untick Master Send**. Record-enable it and turn on input monitoring (speaker icon).
- **T7.** **Time Alignment folder.** Create tracks for FOH L, FOH R and Subs, plus Fill L and Fill R [template]. **Fill L / Fill R are the delay clusters** (called Delay Fill in Step 0 and CalTemp). Drag them all into a folder track named Time Alignment. Open its routing and untick Master Send.
- **T8.** **FFT view.** Add ReaEQ through the FX button on Tones and on RTA, so you can watch the spectrum. [template: ReaEQ is also on Playback, bypassed]
- **T9.** **Save as a project template** and name it. Leave record inputs unassigned in the template; assign the mic in step 8 below, per venue.

⚠ *The template has no phase correction plug-in. Add one to the mix bus by hand (step 2 below). Muted generators are intentional: nothing plays into the PA until you unmute one on purpose.*

A. Set up the DAW (once)

1. Open the DAW template built above (e.g., Reaper)
2. Insert a phase correction plug-in into the mix bus
3. Set the RTA microphone at the sweet spot / overlap reference position — the mix position or the middle of the audience area, the same single reference for every pass (a system cannot be aligned from several reference points at once)
4. Create a record track for the main speaker (Speaker A)
5. Create a record track for the delay / fill speaker (Speaker B)
6. Disable the RTA output routing, so the mic is never sent to the speakers
7. Create a playback track with a full-range transient clip — snare for mid/high-cluster alignment, kick for sub-to-main crossover alignment
8. Assign the RTA mic channel to all record inputs

B. Record each pass

9. Solo the playback on the main speaker only

10. Record a 10-second pass for Speaker A (Main) on its own DAW track
11. Solo the playback on the delay / fill speaker only
12. Record a 10-second pass for Speaker B (Delay/Fill) on its own DAW track

C. Match the recordings

13. Zoom both recorded tracks to the first transient and stack them
14. Nudge waveforms in the DAW to align the leading positive-compression edge for minimum phase inversion, cycle by cycle
15. Check the time difference between the two recorded tracks
16. Note the resulting delay offset (ms)
17. CONFIRM this offset against a live dual-channel FFT transfer-function trace before committing it
18. Apply the confirmed delay offset directly into the speaker management DSP, on the delay speaker's output

⚠ *Waveform nudging is a first-pass estimate only. Early reflections can fool it, and it doesn't guarantee full-band phase coherence — always validate against the FFT trace in Step 3 before writing the value to DSP.*

Step 3 — Live Transfer Function Tuning

1. Set up dual-channel FFT routing in the DAW: Channel 1 = clean reference loop (direct output), Channel 2 = measurement mic
2. Verify the live FFT comparison using pink noise, sine sweeps, AND live music
3. Observe the magnitude delta between reference and measured signal
4. If band variance is within ±3–6 dB → accept the curve
5. If band variance exceeds 10 dB → STOP. Reposition or re-aim the speaker physically. Do not EQ around it.
6. Apply only gentle, subtractive, minimum-phase EQ on genuine peaks
7. NEVER boost a null — if it's caused by destructive-phase cancellation, boosting raises the direct and reflected wave equally, doing nothing but consuming headroom and risking driver excursion
8. Maintain at least 40 dB SNR between the test signal and the ambient noise floor for any measurement to be valid

Step 4 — Playback & Coverage Verification

1. Perform a subjective playback check using reference multi-track audio recordings
2. Walk the venue and map SPL distribution against the coverage targets (front/middle/back rows above)
3. If the middle row drops more than 10 dB, deploy time-aligned delay fills rather than pushing FOH volume
4. Confirm final delay values are correctly stored in the speaker management DSP

Pre-Show Sign-Off

☐	Phase	Criteria
☐	1. Power & SPL	Ambient measured; unity pink noise at +40 dB over ambient; faders at −20 dB nominal
☐	2. Time Alignment	Transients recorded per zone; waveform-nudged; delay confirmed via FFT; stored in DSP
☐	3. Transfer Function	Dual-channel FFT active; variance ±3–6 dB; no boosted nulls; repositioning done if >10 dB
☐	4. Coverage Audit	Front ≤100 dB; mid drop ≤6 dB; delay fills active if mid/back drop >10 dB

⚠ *This sequence is airtight and logical: each step validates the physical preconditions required for the next. Never skip ahead to EQ (Step 3) before SPL (Step 1) and time alignment (Step 2) are confirmed.*

Full reasoning, McCarthy/SIM methodology background, and worked examples live in the main AudioNinjaScrolls manuscript — this is the quick-reference workflow companion.